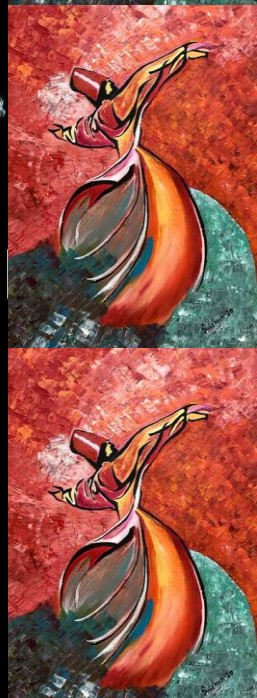




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



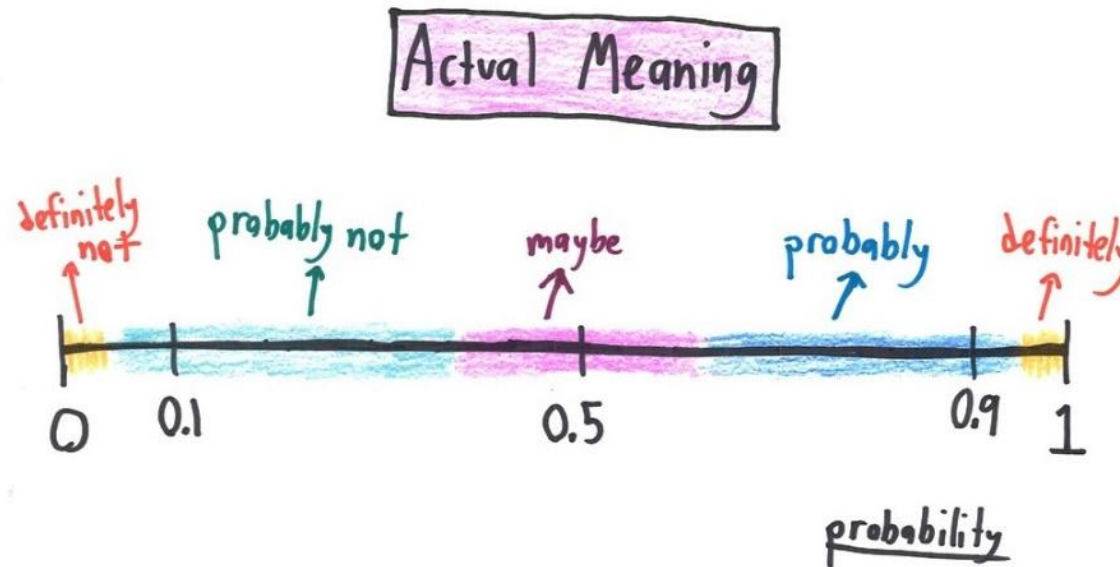


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 2 - Lecture No 03 – 27th August 2025



Announcements!

1. Lectures up to No 02 posted
2. Supplemental Reading on set theory posted (for Unit No 1) – please go through it
3. Quiz No 01 – next week on September 01, 2025 (Wednesday) from Unit No 01

Agenda for today

1. Unit 1: Probability models and axioms (recap)
2. Probability calculation steps
 - Some examples

Unit 1: Probability models and axioms Lecture outline

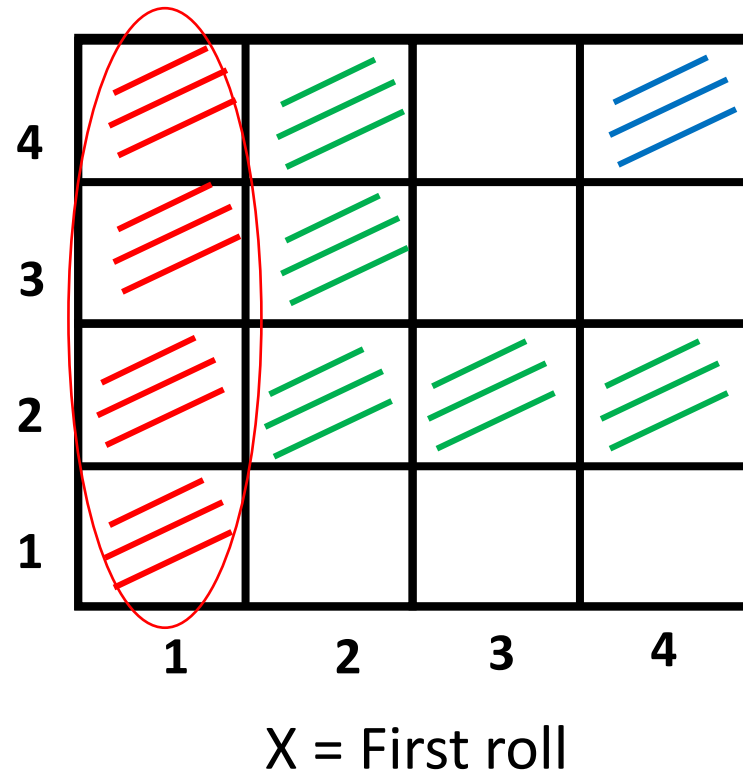
- Sample space
- Probability laws
 - Axioms
 - Some properties
- Examples
 - Discrete
 - Continuous
- Interpretations of probabilities

Probability calculation steps (sequence of 4 steps)

1. Specify the sample space
2. Specify the probability law
3. Identify an event of interest
4. Calculate...

Probability calculation: discrete/finite example

- Two rolls of a tetrahedral die



- Let every possible outcome have probability $1/16$

- $P(X = 1) = 4 * \frac{1}{16} = \frac{1}{4}$

Let $Z = \min(X, Y)$

$$X=2, Y=3, \rightarrow Z=2$$

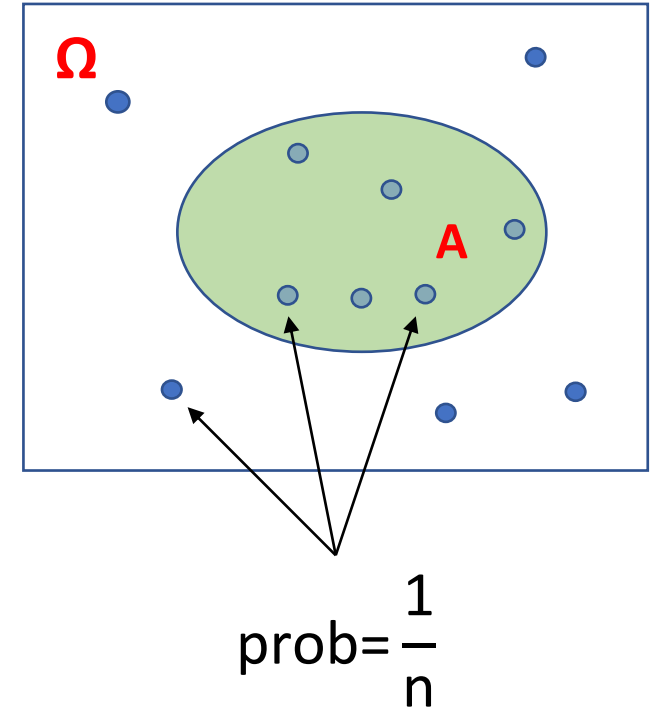
- $P(Z = 4) = \frac{1}{16}$

- $P(Z = 2) = 5 * \frac{1}{16}$

Discrete uniform law

- Assume Ω consists of n equally likely elements
- Assume A consists of k elements

$$P(A) = k * \frac{1}{n}$$



Probability calculation steps (sequence of 4 steps)

1. Specify the sample space
2. Specify the probability law
3. Identify an event of interest
4. Calculate...

Probability calculation: discrete but infinite sample space

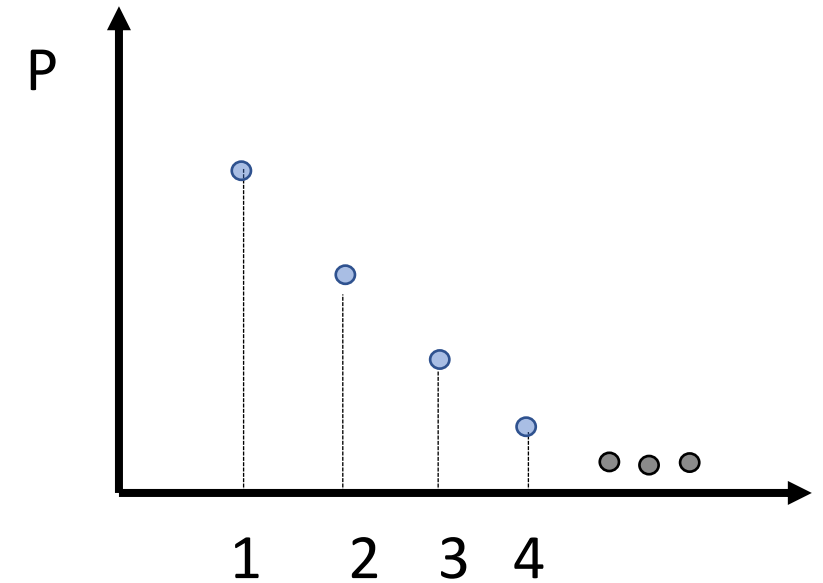
Specify the sample space

1. Sample space: $\{1, 2, \dots\}$

Specify the probability law

2. We are given $P(n) = \frac{1}{2^n}$, $n=1, 2, \dots$

$$\rightarrow \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{2} * \frac{1}{1 - \frac{1}{2}} = 1$$



Identify an event of interest

3. $P(\text{outcome is even}) = P(\{2, 4, 6, \dots\})$

Calculate the probability

$$= P(\{2\} \cup \{4\} \cup \{6\} \cup \dots) = P(2) + P(4) + P(6) + \dots$$

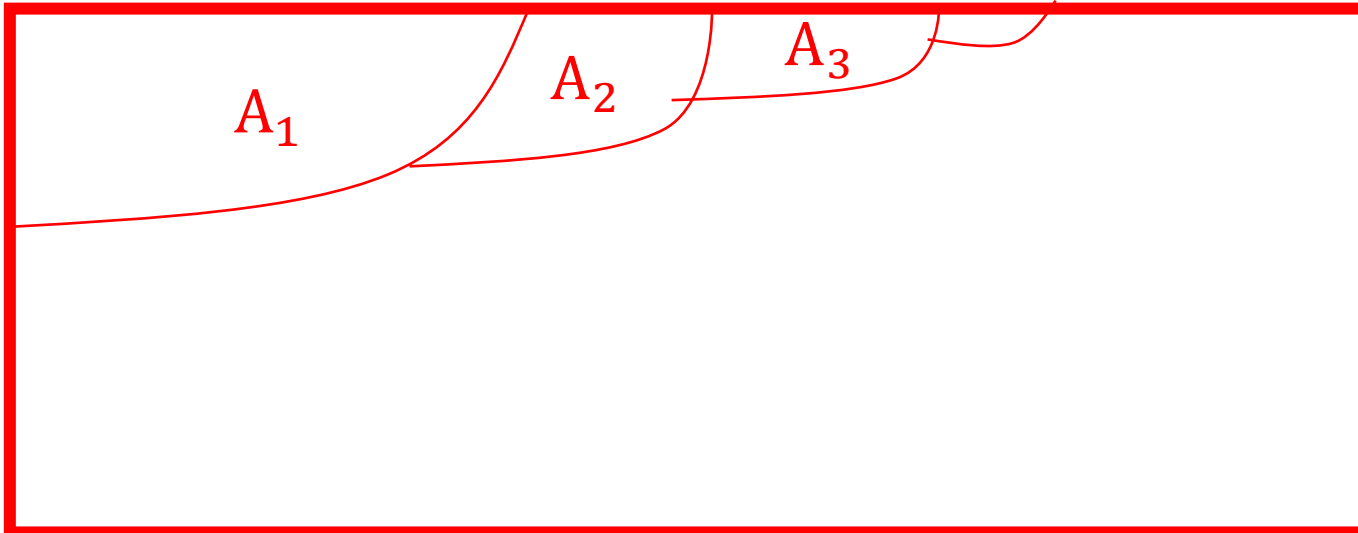
$$= \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \dots = \frac{1}{4} \left(1 + \frac{1}{4} + \frac{1}{4^2} + \dots \right) = \frac{1}{4} * \frac{1}{1 - \frac{1}{4}} = \frac{1}{3}$$

Countable additivity axiom

- Strengthens the finite additivity axiom

Countable Additivity Axiom:

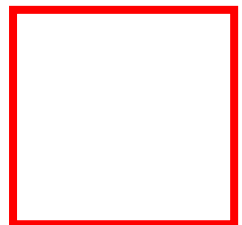
If $A_1, A_2, A_3, \dots$ is an infinite sequence of **disjoint** events,
then $P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$



Countable additivity axiom

Countable Additivity Axiom:

If $A_1, A_2, A_3, \dots$ is an infinite sequence of **disjoint** events,
then $P(A_1 \cup A_2 \cup A_3 \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots$

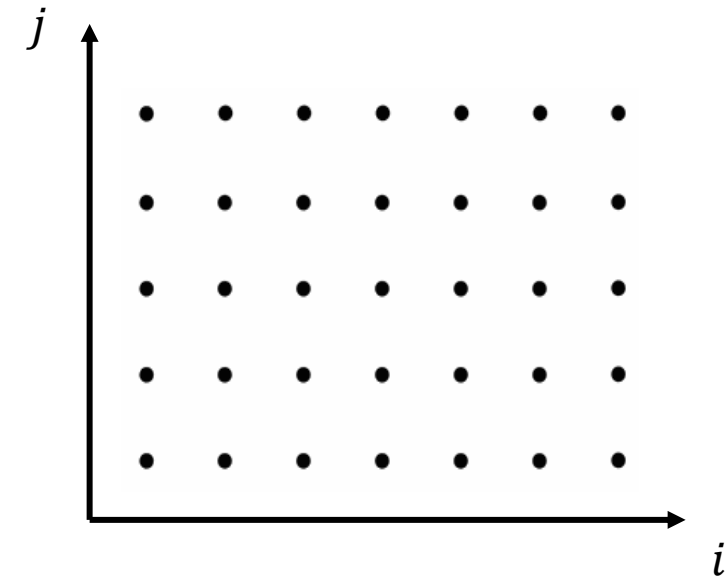
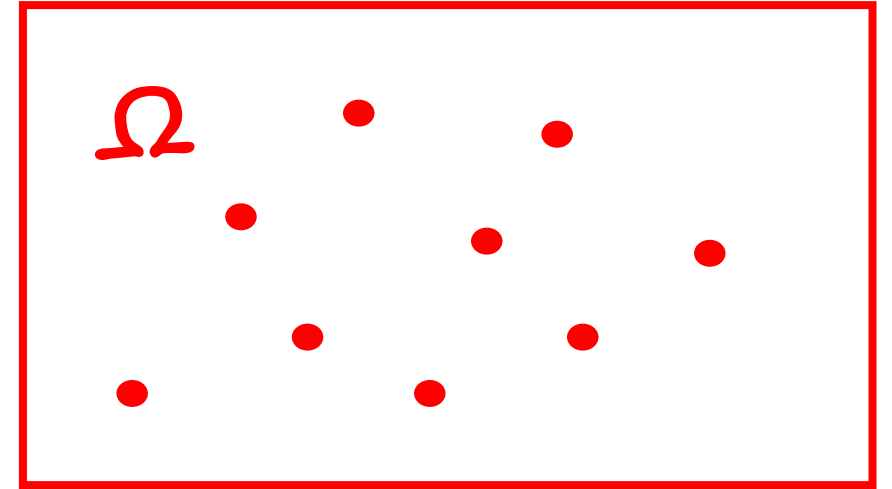


$$1 = P(\Omega) = P\left(\bigcup \{(x, y)\}\right) \stackrel{?}{=} \sum P(\{(x, y)\}) = \sum 0 = 0$$

- Additivity holds only for “countable” sequences of events
- The unit square (similarly, the real line, etc.) is **not countable** (its elements cannot be arranged in a sequence)
- “Area” is a legitimate probability law on the unit square, as long as we do not try to assign probabilities/areas to “very strange” sets

Countable versus uncountable sets

- Countable – can be put in a 1-1 correspondence with positive integers
 - Positive integers?
 - All integers?
 - Pair of integers?
- Uncountable –
 - Unit interval $[0, 1]$



Unit 1: Mathematical Background

- Sets & De Morgan's law
- Infinite Geometric series
- Sums with multiple indices
- Countable & uncountable sets
- Bonferroni inequality

Sets

- A collection of distinct elements

$\{a, b, c, d\}$ *finite*

$\mathbb{R}$: real numbers *infinite*

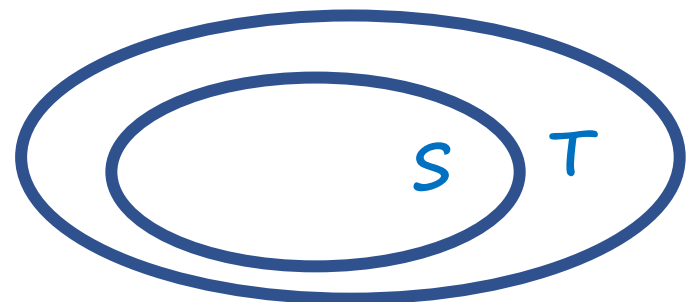
$\{x \in \mathbb{R} : \cos(x) > 1/2\}$

Ω = universal set

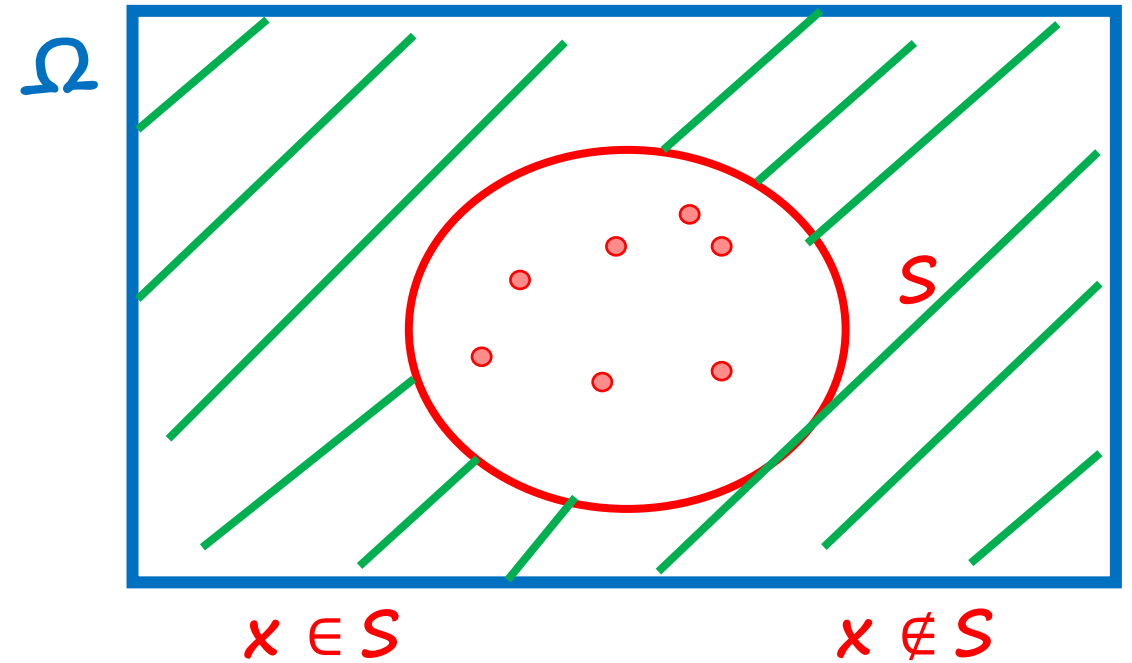
$\emptyset$ = empty set

$$\Omega^c = \emptyset$$

$$(S^c)^c = S$$

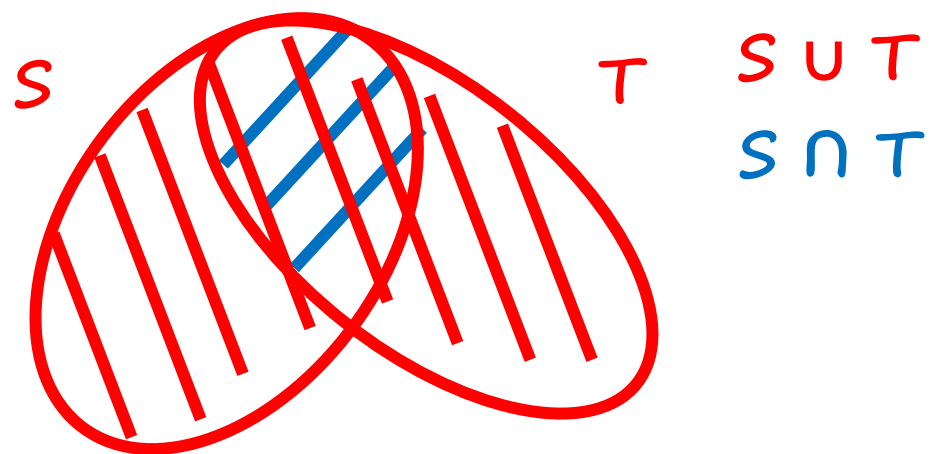


$$S \subset T : x \in S \Rightarrow x \in T$$
$$\subseteq$$



S^c $x \in S^c$ if $x \in \Omega$,
 $x \notin S$

Unions and intersections

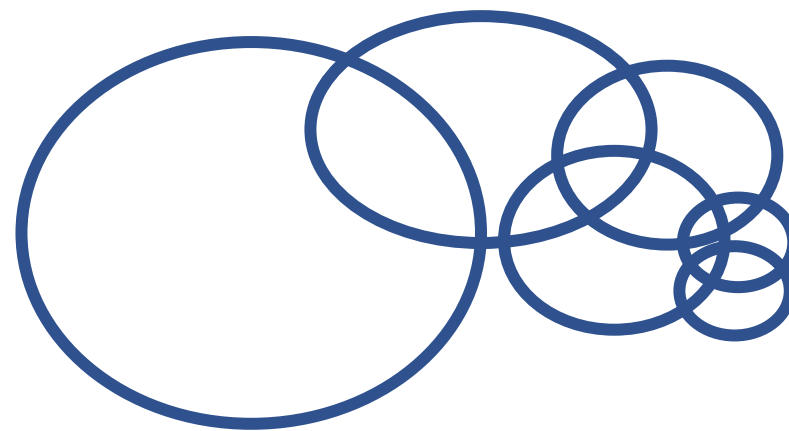


$$x \in S \cup T \Leftrightarrow x \in S \text{ or } x \in T$$
$$x \in S \cap T \Leftrightarrow x \in S \text{ and } x \in T$$

$$S_n \quad n=1, 2, \dots$$

$$x \in \bigcup_n S_n \text{ iff } x \in S_n, \text{ for some } n$$

$$x \in \bigcap_n S_n \text{ iff } x \in S_n, \text{ for all } n$$



Set properties

→ $S \cup T = T \cup S$

→ $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$

→ $(S^c)^c = S,$

$S \cup \Omega = \Omega,$

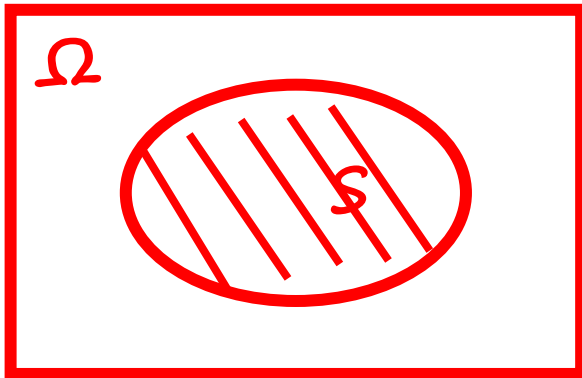
$$S \cup T \cup U$$

$$S \cup (T \cup U) = (S \cup T) \cup U$$

→ $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$

$$S \cap S^c = \emptyset,$$

$$S \cap \Omega = S.$$

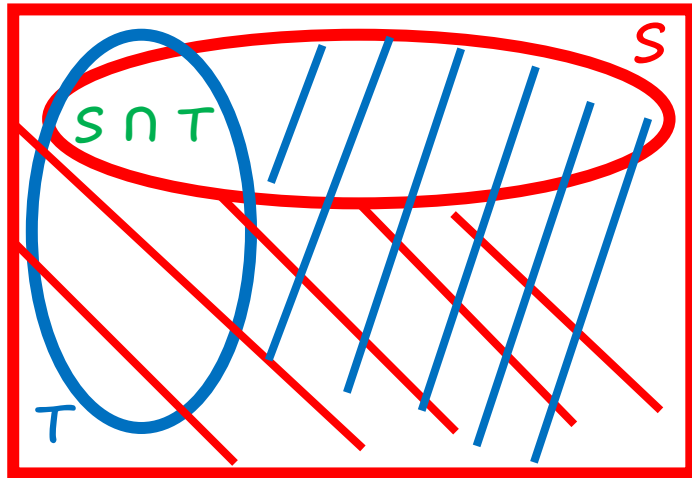


$$S \cap (T \cap U) = (S \cap T) \cap U$$

$$\left. \begin{array}{l} S \subset T \\ T \subset S \end{array} \right\} \Rightarrow S = T$$

De Morgan's laws

$$(S \cap T)^c = S^c \cup T^c$$



$$S \cap (T \cap U) = (S \cap T) \cap U$$

$$\begin{aligned} S &\rightarrow S^c & T &\rightarrow T^c \\ S^c &\rightarrow S & T^c &\rightarrow T \end{aligned}$$

$$(S^c \cap T^c)^c = S \cup T$$

$$S^c \cap T^c = (S \cup T)^c$$

$$\left(\bigcap_n S_n \right)^c = \bigcup_n S_n^c$$

$$\left(\bigcup_n S_n \right)^c = \bigcap_n S_n^c$$

$$x \in (S \cap T)^c \Leftrightarrow x \notin S \cap T \Leftrightarrow \begin{cases} x \notin S \\ \text{or} \\ x \notin T \end{cases} \Leftrightarrow \begin{cases} x \in S^c \\ \text{or} \\ x \in T^c \end{cases} \Leftrightarrow x \in S^c \cup T^c$$

Example - Geniuses + Chocolates

$$P(G) = 0.6 \quad P(C) = 0.7 \quad P(G \cap C) = 0.4$$

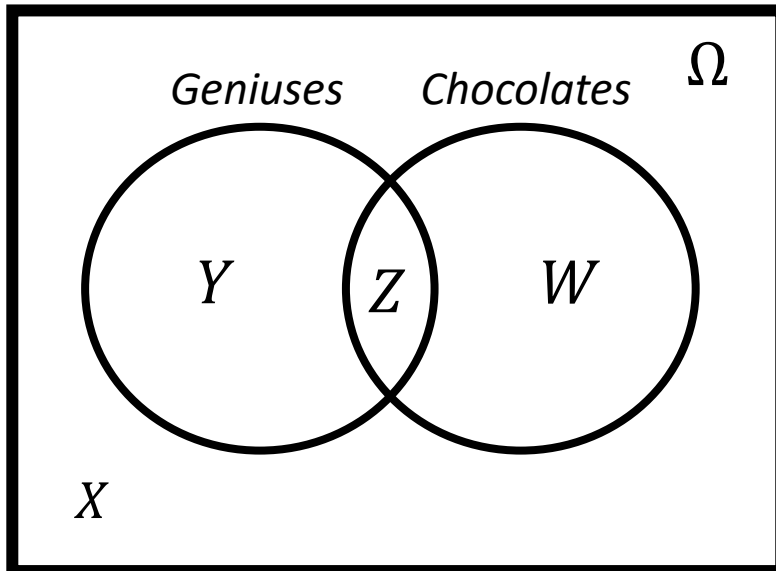
Probability that student picked up at random neither likes chocolate nor is a genius?

$$P(G) = P(Y \cup Z) = P(Y) + P(Z) = 0.6 \Rightarrow P(Y) = 0.2$$

$$P(C) = P(Z \cup W) = P(Z) + P(W) = 0.7 \Rightarrow P(W) = 0.3$$

$$P(Z) = 0.4$$

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$$\begin{aligned} 1 &= P(\Omega) = P(X) + P(Y) + P(Z) + P(W) \\ &= P(X) + 0.2 + 0.4 + 0.3 \\ &\Rightarrow P(X) = 0.1 \end{aligned}$$

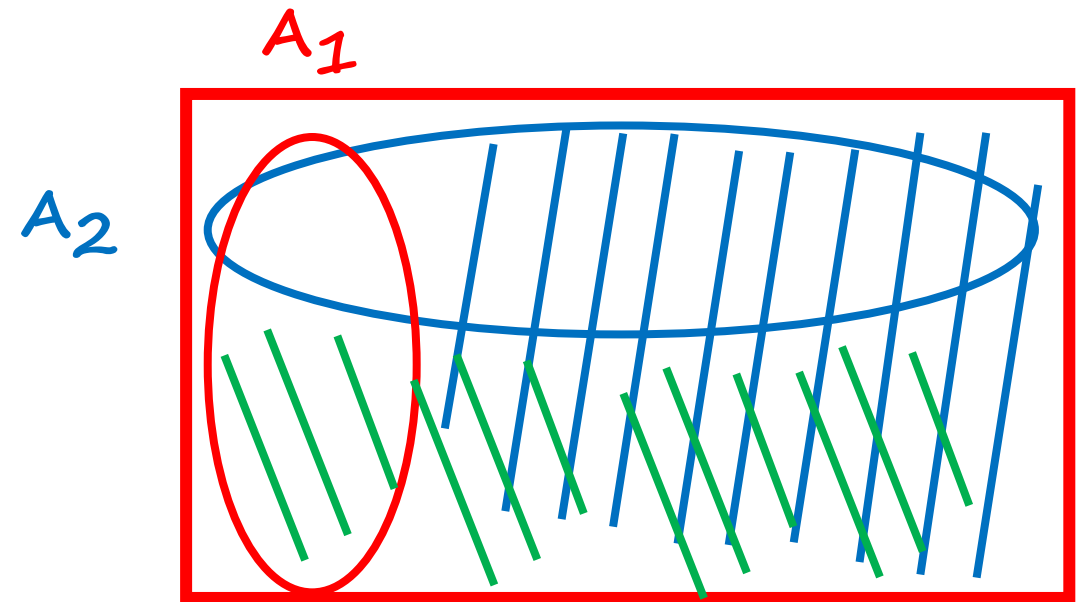
➡ Interpreting the union bound and the Bonferroni inequality

- Suppose that:
 - Very few students are smart A_1
 - Very few students are beautiful A_2
- Then: very few students are smart or beautiful

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

- Suppose that:
 - Most of the students are smart
 - Most students are beautiful
- Then: most students are smart and beautiful

$$P(A_1 \cap A_2) \geq \underline{P(A_1)} + \underline{P(A_2)} - 1$$



The Bonferroni inequality

$$P(A_1 \cap A_2) \geq P(A_1) + P(A_2) - 1$$

$$P(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

$$P(A_1 \cap A_2)^c = P(A_1^c \cup A_2^c) \leq P(A_1^c) + P(A_2^c)$$

$$\begin{aligned} &\stackrel{\parallel}{=} 1 - P(A_1 \cap A_2) \\ &\leq 1 - P(A_1) + 1 - P(A_2) \end{aligned}$$

$$\underline{P(A_1 \cap \dots \cap A_n)} \geq \underline{P(A_1) + \dots + P(A_n)} - \underline{(n - 1)}$$

$$P(A_1 \cap \dots \cap A_n)^c = P(A_1^c \cup \dots \cup A_n^c) \leq P(A_1^c) + \dots + P(A_n^c)$$

$$1 - P(A_1 \cap \dots \cap A_n) \leq (1 - P(A_1)) + \dots + (1 - P(A_n))$$

Interpretations of probability theory

- A narrow view: a branch of math

- Axioms $\rightarrow$ theorems

“Thm:” “Frequency” of event A “is” $P(A)$

- Are the probabilities frequencies?

- $P(\text{coin toss yields heads}) = \frac{1}{2}$

- $P(\text{the prime Minister, Imran Khan will be reelected}) = 0.7$

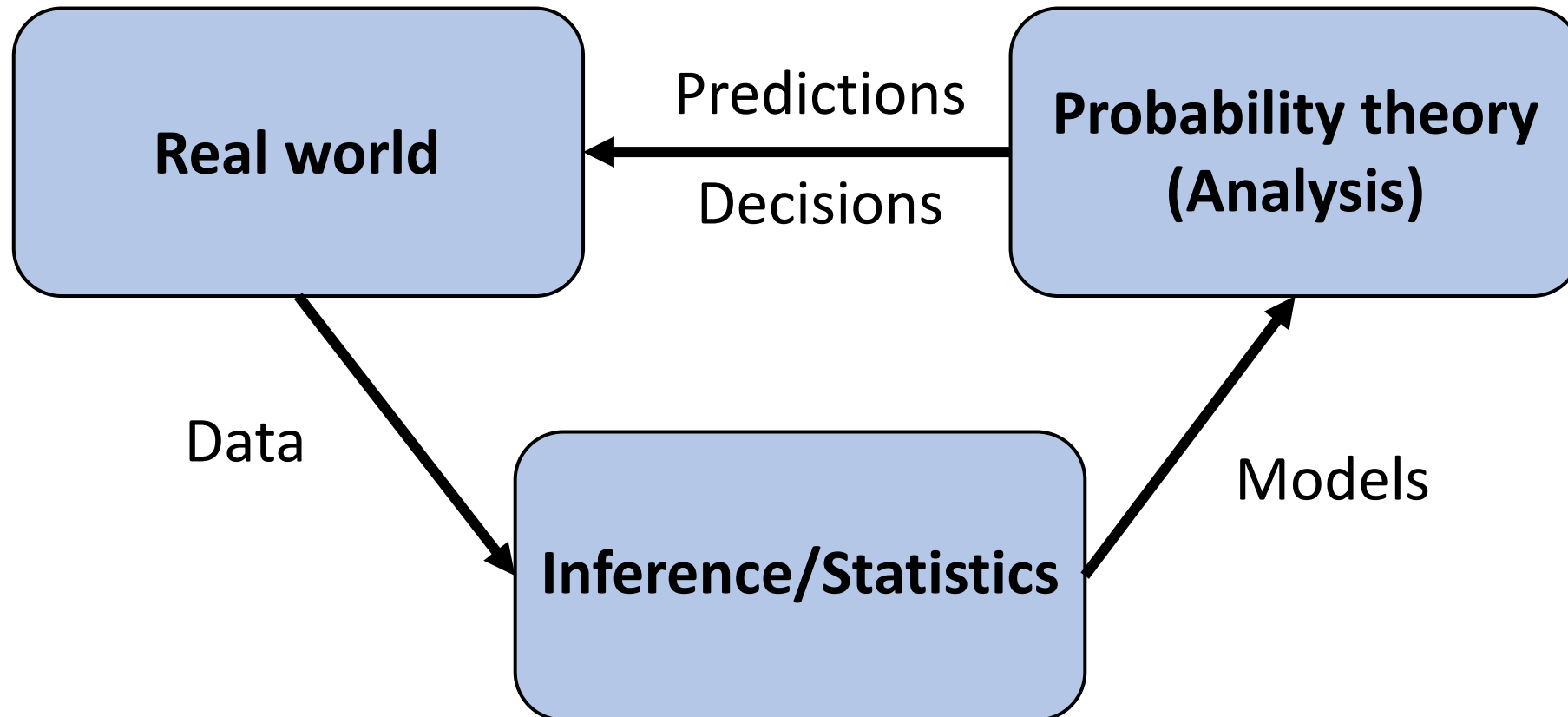
- Probabilities are often interpreted as:

- Description of beliefs

- Betting preferences

Interpretations of probability theory

- A framework for analyzing phenomena with uncertain outcomes
 - Rules for consistent reasoning
 - Used predictions and decisions



Some mathematical concepts
(Self-study)

Mathematical background: Geometric Series

$$S = \sum_{i=0}^{\infty} \alpha^i = 1 + \alpha + \alpha^2 + \dots = \frac{1}{1-\alpha} \quad |\alpha| < 1$$

$$(1 - \alpha)(1 + \alpha + \dots + \alpha^n) = 1 - \alpha^{n+1}$$

$$n \rightarrow \infty$$

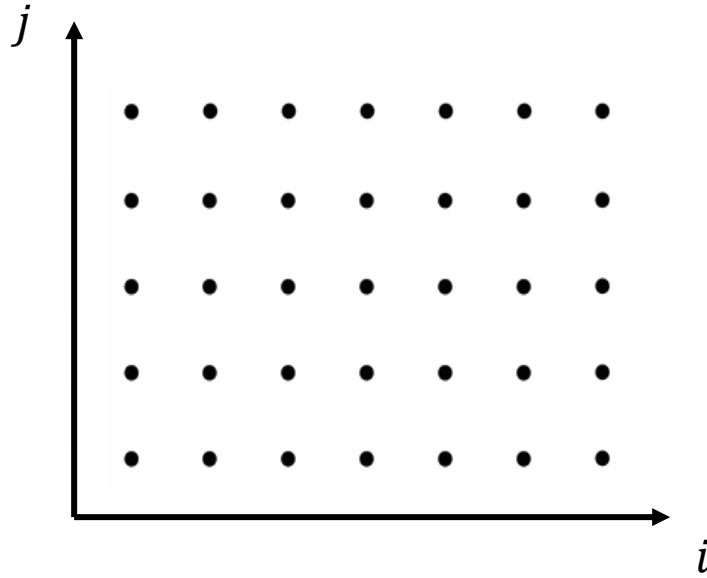
$$(1 - \alpha)S = 1$$

$$S = 1 + \sum_{i=1}^{\infty} \alpha^i = 1 + \alpha \sum_{i=0}^{\infty} \alpha^i = 1 + \alpha S \Rightarrow S(1 - \alpha) = 1$$

$S < \infty$ taken for granted

About the order of summation in series with multiple indices

$$\sum_{i \geq 1, j \geq 1} a_{ij}$$



$n \rightarrow \infty$

(a) Fix i and let $j \Rightarrow 1 - \infty$

$$\sum_{i=1}^{\infty} \left(\sum_{j=1}^{\infty} a_{ij} \right)$$

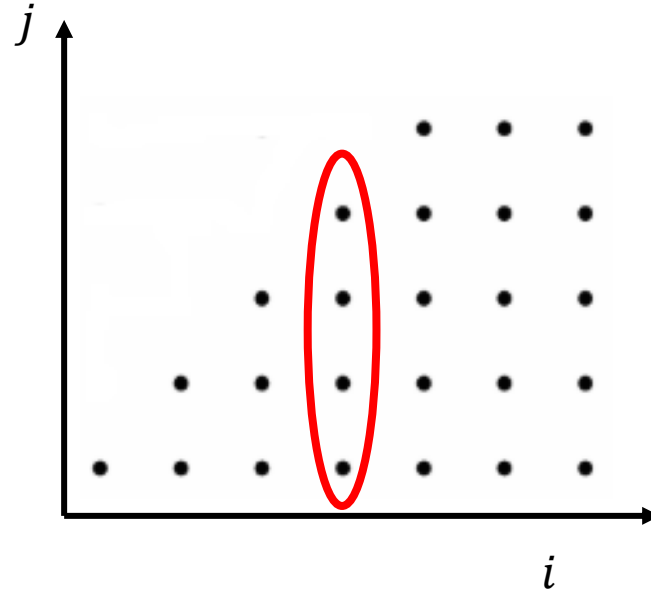
(b) Fix j and let $i \Rightarrow 1 - \infty$

$$\sum_{j=1}^{\infty} \left(\sum_{i=1}^{\infty} a_{ij} \right)$$

if $\sum |a_{ij}| < \infty$, then order does not matter

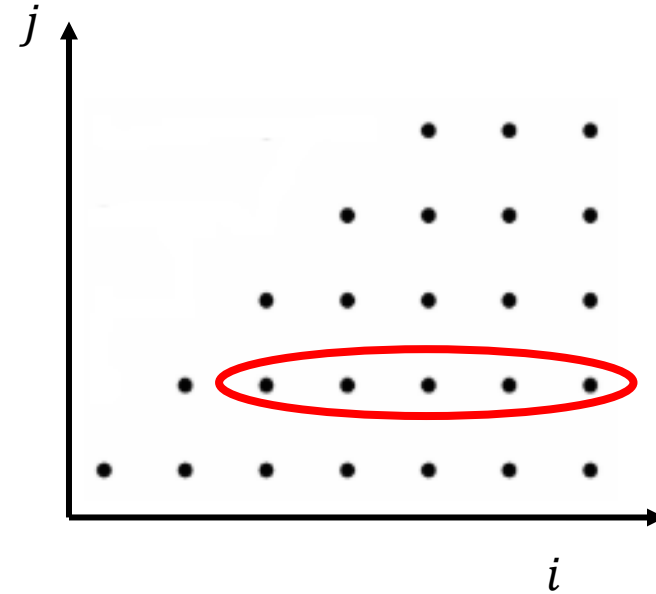
About the order of summation in series with multiple indices

$$\sum_{(i,j): j \leq i}^{\text{if}} |a_{ij}| < \infty$$



$$\sum_{i=1}^{\infty} \sum_{j=1}^i a_{ij}$$

=

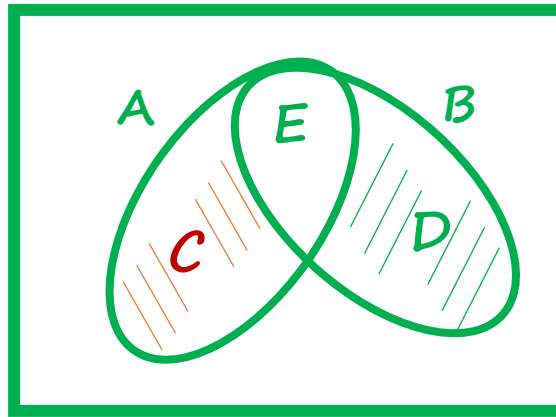


$$\sum_{j=1}^{\infty} \sum_{i=j}^{\infty} a_{ij}$$

4 Practise Problems

Practice Problem 1

Show $P(C \cup D) = P(A) + P(B) - 2P(E)$



$$\begin{aligned} P(C \cup D) &= P(C) + P(D) \\ &= P(C) + P(E) + P(D) + P(E) - 2P(E) \\ &= P(A) + P(B) - 2P(E) \end{aligned}$$

Practice Problem 2

Tim has a peculiar four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to result of the dice.

(a) What is the probability of the number being even?

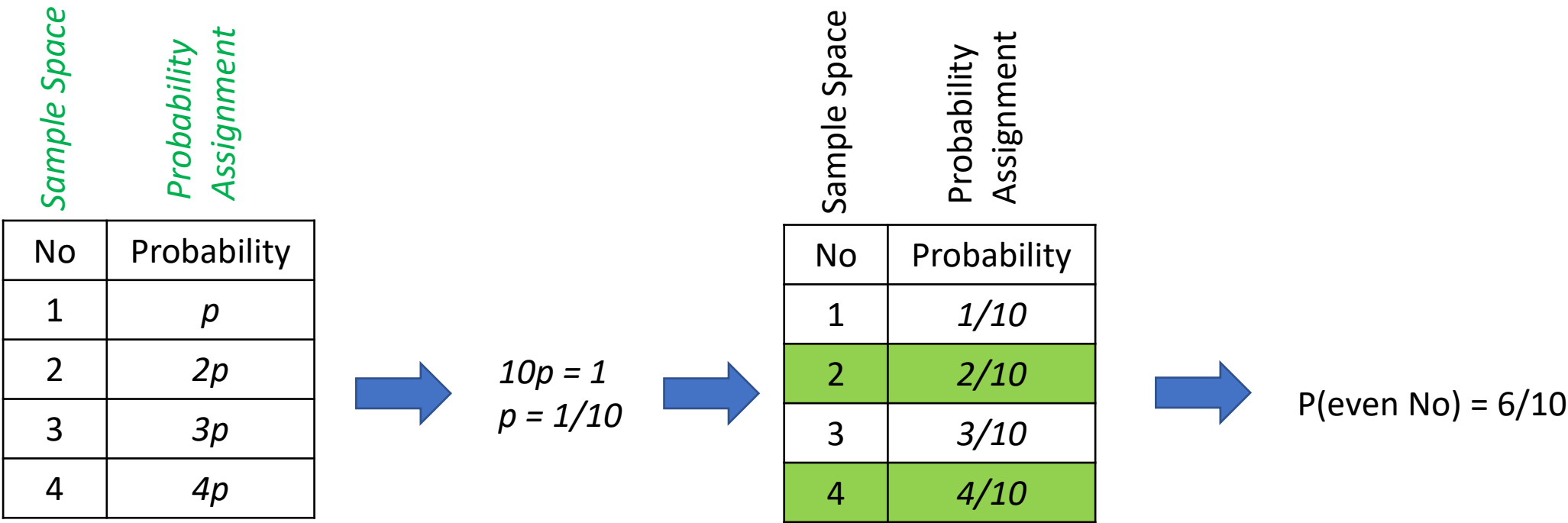
Sample Space

Probability
Assignment

Practice Problem 2

Tim has a peculiar four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to result of the dice.

(a) What is the probability of the number being even?



Practice Problem 3 – *do it in pairs*

Hasan has a peculiar pair of four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to the sum of the results of the two individual dice.

- (a) What is the probability of the sum being even?
- (b) What is the probability of Hasan rolling a 2 and a 3, in any order?

Practice Problem 3 - Solution

Hasan has a peculiar pair of four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to the sum of the results of the two individual dice.

- (a) What is the probability of the sum being even?
- (b) What is the probability of Hasan rolling a 2 and a 3, in any order?

Die 1	Die 2	Sum	P(Sum)
1	1	2	2p
1	2	3	3p
1	3	4	4p
1	4	5	5p
2	1	3	3p
2	2	4	4p
2	3	5	5p
2	4	6	6p
3	1	4	4p
3	2	5	5p
3	3	6	6p
3	4	7	7p
4	1	5	5p
4	2	6	6p
4	3	7	7p
4	4	8	8p
		Total	80p

(a)

$$P(\text{Even sum}) = 2p + 4p + 4p + 6p + 4p + 6p + 6p + 8p = 40p = \boxed{1/2}$$

(b)

$$P(\text{Rolling a 2 and a 3}) = P(2, 3) + P(3, 2) = 5p + 5p = 10p = \boxed{1/8}$$

$$P(\text{All outcomes}) = 80p \text{ (Total from the table)}$$

and therefore

$$p = \frac{1}{80}$$